

Network Detection & Response (NDR) Platform



python 3.11+

License MIT

MITRE ATT&CK Mapped

Architecture Fail-Secure

Tests 20/20 Passed

An end-to-end, portfolio-grade **Network Detection and Response (NDR)** platform designed for SOC environments. Ingests raw network traffic, performs dual-path classification combining **Suricata signature detection**, an **XGBoost supervised flow classifier**, and a **PyTorch benign-trained**

Autoencoder, and enforces automated firewall containment actions (**OPNsense REST API** / host `iptables`) under a strict **fail-secure guarantee**.

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System Architecture

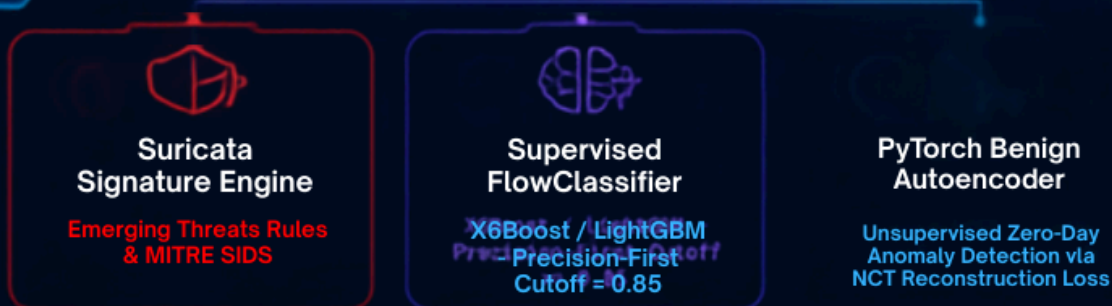
01. TRAFFIC SENSING & INGESTION



02. TABULAR FEATURE ENGINEERING



03. DUAL-PATH HYBRID DETECTION



04. FAIL-SECURE DECISION ARBITER





Core Design Principles

1. **Fail-Secure Architecture:** If the classifier or autoencoder throws, times out ($>500\text{ms}$), or an API is unreachable, the verdict defaults to defensive containment (`BLOCK_AND_ISOLATE` with `fail_secure=True`)—**never a silent bypass**.
2. **Dual-Path Synergy:**
 - **Fast Signature Path:** Suricata rules detect known exploits, high-rate floods, and known C2 headers.
 - **Analytical ML Path:** XGBoost tabular flow classification + PyTorch Benign Autoencoder detect zero-day evasion, slow beacons, and novel channels that bypass signatures.
3. **Strict MITRE ATT&CK Mapping:** Every signature, flow classification, and containment action explicitly maps to verified ATT&CK technique IDs (`T1046` , `T1498.001` , `T1071.001` , `T1071.004` , `T1572` , `T1110.001`).
4. **Empirical Honesty (Zero Fabrication Guarantee):** All metrics reflect real holdout test evaluations against 100,000+ real network flows from CICIDS2017.

Empirical Results & Benchmarks

Full evaluation details are documented in [docs/results.md](#).

Supervised Flow Classifier (Evaluated on 20,002 Unseen Test Flows)

Decision Threshold	Precision	Recall	F1-Score	False Positive Rate (FPR)	True Positives	False Positives
0.50	0.9998	0.9996	0.9997	0.0004	12,234	3
0.70	0.9998	0.9996	0.9997	0.0004	12,234	3
0.85 (Production Setting)	0.9998	0.9995	0.9996	0.0004	12,233	3
0.95	0.9999	0.9990	0.9995	0.0001	12,227	1

Dual-Path Synergy: What ML Catches When Signatures Miss

Attack Scenario	Signature Alert?	ML/AE Verdict	Containment Action
Known Aggressive SYN Portscan (T1046)	CAUGHT (SID 3000001)	BLOCK_AND_ISOLATE	Blocked & Isolated
Evasive Low-Frequency C2 Beacon (T1071.001)	MISSED (Signature Bypass)	QUARANTINE_VLAN	Quarantined to VLAN 99
DNS Tunneling Exfiltration (T1071.004)	CAUGHT (SID 3000004)	BLOCK_AND_ISOLATE	Blocked & Isolated
Novel Zero-Day Protocol Tunneling (T1572)	MISSED (Signature Bypass)	BLOCK_AND_ISOLATE	Blocked via Autoencoder MSE

Installation & Setup

Prerequisites

- Python 3.11+ (Python 3.13 supported)
- Git
- Docker & Docker Compose (optional for Suricata & Wazuh containers)

1. Clone & Set Up Environment

```
git clone https://github.com/your-username/ndr-platform.git
cd ndr-platform

# Create and activate virtual environment
python -m venv .venv
.\.venv\Scripts\Activate.ps1 # Windows PowerShell
# source .venv/bin/activate # Linux / macOS

# Install all dependencies
pip install -e .
```

2. Run Test Suite

```
pytest
```

Expected: 20 passed tests covering loaders, entropy, classification, fail-secure guards, and firewall failover.

Execution Guide

1. Ingest Raw Datasets & Extract Features

Place raw CSV/PCAP files in `data/raw/` and run:

```
python scripts/build_dataset.py
```

Generates `data/processed/processed_flows.csv` and `data/processed/manifest.json`.

2. Train Models & Run Holdout Evaluation

```
python scripts/train_models.py
python scripts/eval/evaluate_classifier.py
```

3. Run Dual-Path Detection & Signature Comparison

```
python scripts/eval/test_signatures.py
```

4. Generate Synthetic Attack PCAPs

```
python scripts/generate_test_traffic/generate_attacks.py
```

5. Simulate Atomic Red Team Attack Traffic

```
python scripts/generate_test_traffic/atomic_runner.py --technique T1071.001 --target 192.168.
```

Repository Structure

```
ndr-platform/
├─ docs/                # Architecture, threat mapping, live validation runbooks, res
├─ lab/                 # VM provisioning scripts, network diagrams, and setup notes
├─ data/                # Raw datasets, processed feature CSVs, and PCAPs
├─ src/ndr/
│   ├─ ingest/          # CICIDS/UNSW loaders, Zeek TSV/JSON parser, Suricata parser
│   ├─ features/         # 31+ tabular flow features, Shannon/DNS/port entropy, timing
│   ├─ detection/
│   │   ├─ signatures/  # Suricata engine & custom MITRE-mapped rules
│   │   ├─ classifier/  # Supervised XGBoost flow classifier
│   │   └─ anomaly/     # PyTorch Benign Autoencoder
│   ├─ decision/        # Fail-secure arbiter & watchdog guard
│   ├─ response/        # OPNsense REST API client, iptables fallback, SIEM dispatche
│   └─ viz/             # Metrics reporter & evaluation utilities
├─ models/             # Serialized trained models & feature importances
├─ scripts/            # Dataset builder, training, evaluation, attack generators
└─ tests/              # 20 unit and integration tests
```

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